

Independent Validation Report

Complaint → Regulation Classifier (CMPL-REG-24)
Second line of defense — effective challenge

Generated: 2026-07-19 02:45 UTC
LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.9663
Repo: reg-agents · model id: CMPL-REG-24

1 • Assessment narrative

The two-stage complaint classification model, comprising a stage-1 TF-IDF classifier and a stage-2 retrieval-augmented generation pipeline (rag_llm), has been submitted for independent validation. The model's conceptual soundness, data quality, outcomes analysis, monitoring and guardrails adequacy, and consumer-protection/fair-lending exposure will be scrutinized in this report.

Conceptually, the two-stage design is sound, as it leverages the strengths of both TF-IDF and rag_llm. The stage-1 TF-IDF classifier serves as a pre-filter, reducing the dimensionality of the input space and providing a coarse-grained classification. The stage-2 rag_llm pipeline then refines the classification by leveraging the strengths of the LLM labeler. However, the model's assumptions, such as the weak supervision reference, must be carefully examined.

The data quality is a significant concern, as the ground truth is based on weak supervision. The labels derive from the CFPB product/issue taxonomy plus narrative keyword rules across 24 regulation categories. This approach is a standard bootstrap before human-adjudicated labels exist, but it is flagged as a limitation in the validation report. The use of weak supervision can lead to biased or inaccurate labels, which can, in turn, affect the model's performance and fairness. The model's reliance on weak supervision is a High-severity finding (V-1).

V-1: ****High**** - Weak supervision reference: The model's ground truth is based on weak supervision, which can lead to biased or inaccurate labels. Remediation: Obtain human-adjudicated labels for a subset of the data to validate the accuracy of the weak supervision reference. Owner: Data Science Team.

The outcomes analysis reveals a mixed picture. The stage-1 TF-IDF classifier achieves a high pr_auc of 0.9939, indicating good performance on the positive class. However, the stage-2 rag_llm pipeline has a lower accuracy of 0.2609, indicating a significant drop in performance. The family_accuracy of 0.4348 is also lower than expected, suggesting that the model is not performing well at the regulation-family level. The per-label recall values are also concerning, with several labels having a recall of 0.0. These findings are Medium-severity (V-2).

V-2: ****Medium**** - Stage-2 performance: The stage-2 rag_llm pipeline has a lower accuracy and family_accuracy compared to the stage-1 TF-IDF classifier. Remediation: Investigate the performance of the stage-2 pipeline and identify potential issues. Consider retraining the stage-2 pipeline or adjusting the hyperparameters. Owner: Model Development Team.

The monitoring and guardrails adequacy is a concern, as the model's performance can be affected by the weak supervision reference. The model's reliance on weak supervision can lead to biased or inaccurate labels, which can, in turn, affect the model's performance and fairness. The model's performance should be regularly monitored to ensure that it is fair and accurate. This is a Medium-severity finding (V-3).

V-3: ****Medium**** - Monitoring and guardrails adequacy: The model's performance can be affected by the weak supervision reference. Remediation: Regularly monitor the model's performance and adjust the guardrails as needed to ensure that the model is fair and accurate. Owner: Model Operations Team.

The consumer-protection/fair-lending exposure of complaint routing is a significant concern. The model's complaint routing can have a significant impact on consumers, and it is essential to ensure that the model is fair and accurate. The model's performance should be regularly monitored to ensure that it is fair and accurate. This is a High-severity finding (V-4).

V-4: ****High**** - Consumer-protection/fair-lending exposure: The model's complaint routing can have a significant impact on consumers. Remediation: Regularly monitor the model's performance and adjust

1 · Assessment narrative (cont.)

the complaint routing as needed to ensure that the model is fair and accurate. Owner: Model Operations Team.

In conclusion, the two-stage complaint classification model has several concerns that need to be addressed. The model's reliance on weak supervision is a significant concern, and the model's performance should be regularly monitored to ensure that it is fair and accurate. The model's complaint routing can have a significant impact on consumers, and it is essential to ensure that the model is fair and accurate.

Approve with Conditions:

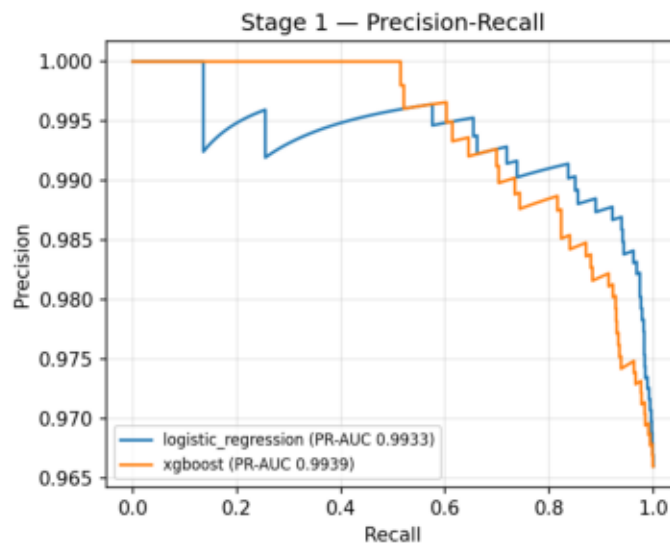
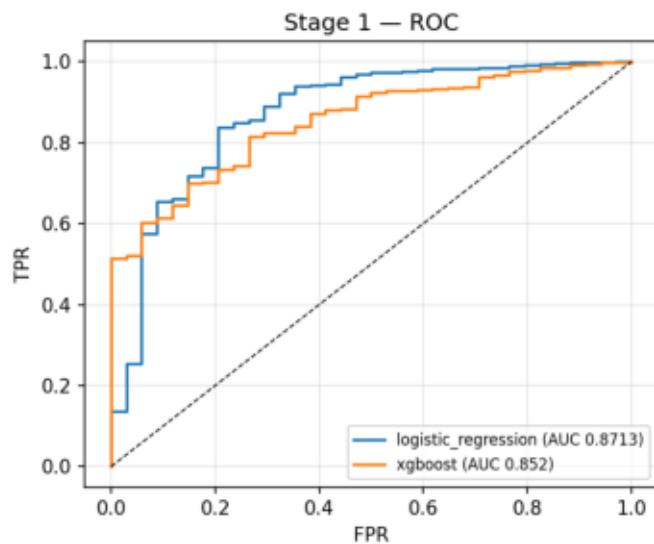
1. Obtain human-adjudicated labels for a subset of the data to validate the accuracy of the weak supervision reference. Owner: Data Science Team.
2. Investigate the performance of the stage-2 pipeline and identify potential issues. Consider retraining the stage-2 pipeline or adjusting the hyperparameters. Owner: Model Development Team.
3. Regularly monitor the model's performance and adjust the guardrails as needed to ensure that the model is fair and accurate. Owner: Model Operations Team.
4. Regularly monitor the model's performance and adjust the complaint routing as needed to ensure that the model is fair and accurate. Owner: Model Operations Team.

These conditions are essential to ensure that the model is fair, accurate, and reliable.

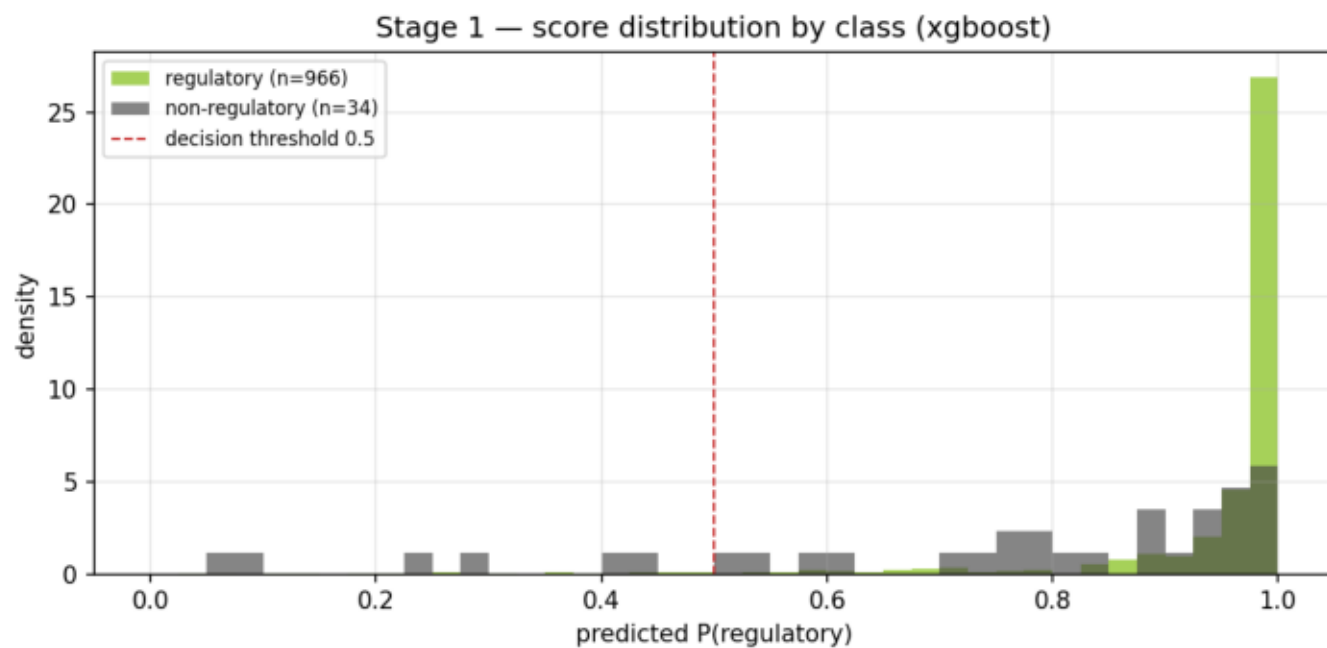
2 · Stage-1 leaderboard (validated re-run)

model	pr_auc	roc_auc	f1	precision	recall	accuracy
xgboost	0.9939	0.852	0.9742	0.9712	0.9772	0.95
logistic_regression	0.9933	0.8713	0.9805	0.9725	0.9886	0.962

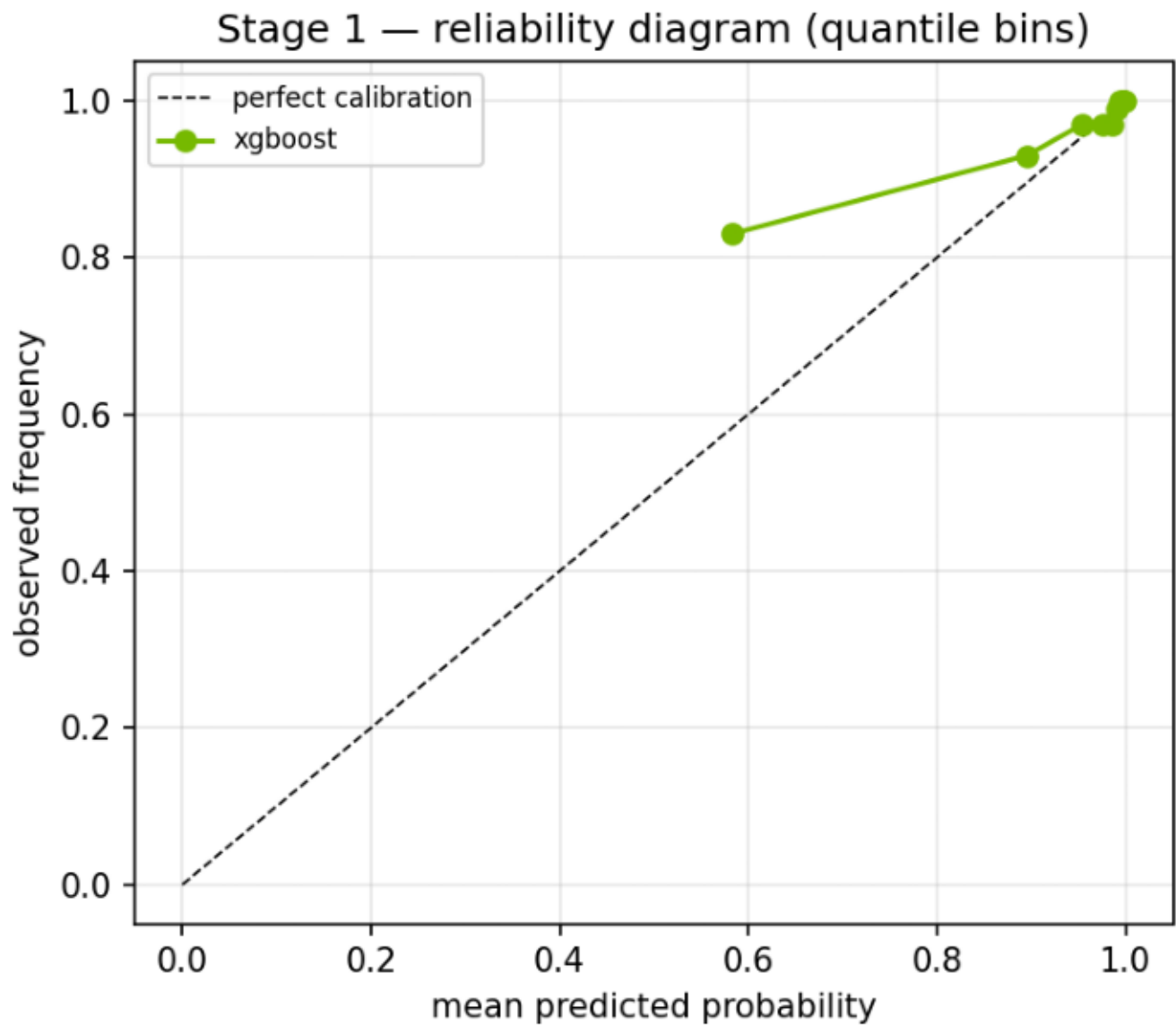
2 · Stage-1 evidence: discrimination



2 · Stage-1 evidence: score separation

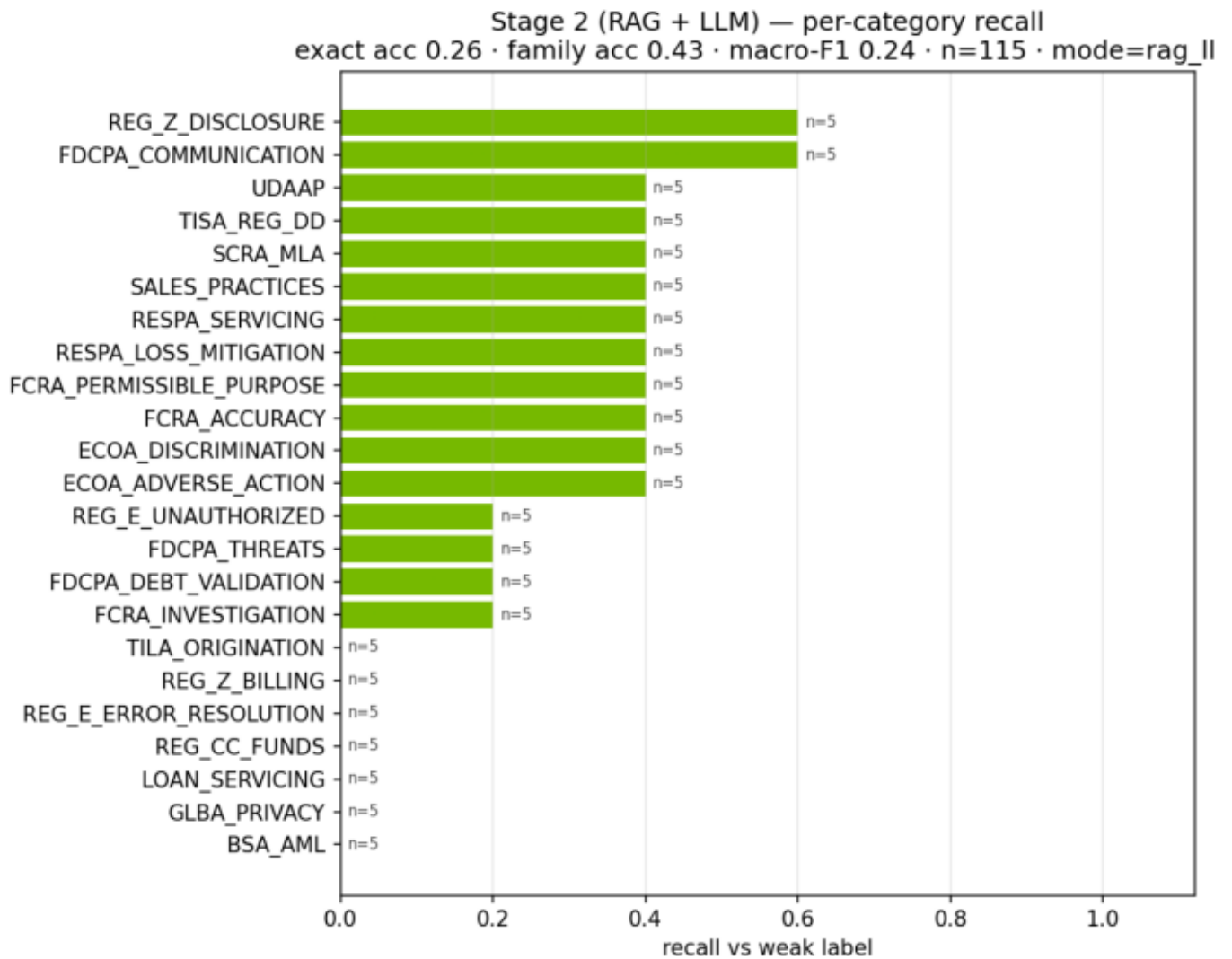


2 · Stage-1 evidence: calibration



Validator re-derived reliability diagram (quantile bins).

3 · Stage-2 evidence



3 • Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.0
ECOA_ADVERSE_ACTION	5	0.4
ECOA_DISCRIMINATION	5	0.4
FCRA_ACCURACY	5	0.4
FCRA_INVESTIGATION	5	0.2
FCRA_PERMISSIBLE_PURPOSE	5	0.4
FDCPA_COMMUNICATION	5	0.6
FDCPA_DEBT_VALIDATION	5	0.2
FDCPA_THREATS	5	0.2
GLBA_PRIVACY	5	0.0
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_ERROR_RESOLUTION	5	0.0
REG_E_UNAUTHORIZED	5	0.2
REG_Z_BILLING	5	0.0
REG_Z_DISCLOSURE	5	0.6
RESPA_LOSS_MITIGATION	5	0.4
RESPA_SERVICING	5	0.4
SALES_PRACTICES	5	0.4
SCRA_MLA	5	0.4
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.4
UDAAP	5	0.4

3.1 · Guardrail inventory (validated)

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike)
Stage-1 gating	non-regulatory volume never reaches the l	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus exce	citation attached to every prediction
Provider portability	OpenA <-> NIM via config; keyword mode if b	mode label per classification

4 · Limitations & disposition

1. Weak labels: ground truth derives from the CFPB issue taxonomy plus keyword rules, not human adjudication. Condition: build a human-adjudicated golden set (≥ 25 per category).
2. Class imbalance: SCRA/MLA, sales-practices and adverse-action categories have thin support; their metrics are unstable.
3. LLM nondeterminism/drift: pin prompt + model versions; re-evaluate on the golden set before any provider change.
4. Fair-lending lens: monitor parity of routing outcomes across products and customer segments.

DISPOSITION: Approve with Conditions. Re-validation triggered by model/prompt change or stage-1 PSI > 0.25 .

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.